Onyx — AppKit reference

Onyx · multi-process OS on Raspberry Pi 4

*The reference of **AppKit** (`SD:/lib/appkit.so`, `user/Kits/appkit`): what it is for, how a program uses it, and every operation it exposes. The operations' part is made from the kit's headers by `tools/docgen/kitdocs.py` — the headers are the source. Overview of all the kits: The Kits.*

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What it is

AppKit is what makes a program run: its one link to the system. Every call a program makes to the kernel is a function of AppKit (`kapi_*`), and AppKit carries the small services every program needs — strings, the console, `.ini` files, the keyboard layout, the starting of programs. Only AppKit reads the kernel's table, so the kernel can change without a program being rebuilt.

Include	`#include "appkit/appkit.h"`
Link	nothing to link: the kernel binds AppKit to every program
Library	`SD:/lib/appkit.so` — 317 entries in its table (`user/Kits/appkit/appkit.abi`, append-only)
Sources	`user/Kits/appkit/`

Using it

AppKit is the program's link to the system: files, processes, time, sockets, the events' pump… (the `kapi_*` calls, listed in the Developer Guide; the windows are UIKit's since 2026-10-08: `uk_win_*`, §4), plus the small services every program needs: strings without a C library, the console, a reader of `.ini` files, the starting of other programs.

A console tool: read a file, print to the console.

```
#include "appkit/appkit.h"

int main (void)
{
    char buf[256];
    void *f = kapi_open ("SD:/etc/autostart");
    if (f == 0) { ax_putln ("no such file"); return 1; }
    int n = kapi_read (f, buf, sizeof buf - 1);
```

```
    kapi_close (f);
    buf[n > 0 ? n : 0] = '\0';

    ax_putln ("SD:/etc/autostart says:");
    ax_putln (buf);
    return 0;
}
```

Settings from an `.ini` file in the program's own folder:

```
if (app_ini_load ("config.ini") >= 0)
{
    const char *name = app_ini_get ("window", "title", "Untitled");
    int width        = app_ini_get_int ("window", "width", 640);
    for (int i = 0; i < app_ini_count (); i++)        // or every entry, in the file's order
        ax_putln (app_ini_key (i));
}
```

Strings (no C library needed):

```
char path[96]; int n = 0;
ax_strcat (path, sizeof path, &n, "SD:/docs/");       // never overflows, always terminated
ax_strcat (path, sizeof path, &n, "notes.txt");
char num[12]; ax_itoa (42, num);
if (ax_streq (num, "42")) ax_putln (path);
```

Starting a program — an application by its name, or a file by whatever runs it (a `.bas` program by BASIC, a game by its emulator):

```
lx_launch ("tinypad", "SD:/docs/notes.txt");          // an app, with its arguments
lx_open ("SD:/games/tetris.gb", 0);                   // a file: by its runner
```

The volumes, USB sticks (kapi v93) — list them, eject a stick, format one:

```
struct kapi_volume v[16];
int n = kapi_vol_list (v, 16, KAPI_VOLS_ROOM);        // SD:, SD1:.., USB1:.., USB1P1:..,
RAM:
for (int i = 0; i < n && i < 16; i++)
    if ((v[i].flags & KAPI_VF_REMOVABLE) && v[i].state == KAPI_VST_MOUNTED)
        ax_putln (v[i].label);                        // a stick plugged in: "USB" + its
label

if (kapi_vol_eject ("USB1:", 0) == -KAPI_EBUSY)       // files open on it (they were synced)
    kapi_vol_eject ("USB1:", KAPI_EJECT_FORCE);       // ... after asking the user

struct kapi_format f = { KAPI_FMT_EXFAT, 0, 0, "PHOTOS" };
kapi_vol_format ("USB1:", &f);                        // erases it; SD: is always refused
```

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Everything the headers declare, in their order — the details are in each header's part below.

Name	What it does	Header
`kapi_abi_version`	The kernel's kapi version (KAPI_ABI_VERSION as the running kernel says it)	`appkit.h`
`kapi_table_slot`	The address of a slot of	`appkit.h`

Name	What it does	Header
	the kernel's table (its index in 8-byte words)	
`kapi_launch`	start the app n (SD:apps/.app/main) as a new process -> 1, 0 failure	`appkit.h`
`kapi_list_tasks`	Every task, one line each " " (state R / S / B / N, kind a = an app or k = a kernel task), into b (s bytes) -> how many.	`appkit.h`
`kapi_kill`	kill the app of that name -> 1, 0 (not running, a kernel task, the caller)	`appkit.h`
`kapi_list_procs`	ps / kill by PID.	`appkit.h`
`kapi_kill_pid`	ps / kill by PID.	`appkit.h`
`kapi_proc_tree`	(v91) A process's tree	`appkit.h`
`kapi_set_keymap`	Keyboard layout	`appkit.h`
`kapi_get_keymap`	Keyboard layout	`appkit.h`
`kapi_exec`	Run an ELF at an absolute path with an argv string (fire-and-forget).	`appkit.h`
`kapi_screen_size`	Framebuffer size in pixels (for edge-pinned borderless windows).	`appkit.h`
`kapi_get_ticks`	Framebuffer size in pixels (for edge-pinned borderless windows).	`appkit.h`
`kapi_msleep`	Framebuffer size in pixels (for edge-pinned borderless windows).	`appkit.h`
`kapi_yield`	Framebuffer size in pixels (for edge-pinned borderless windows).	`appkit.h`
`kapi_exit`	Framebuffer size in pixels (for edge-pinned borderless windows).	`appkit.h`
`kapi_pump_events`	Run what is pending -- the kapi_post calls, then this program's events through their handlers -- and return (no wait).	`appkit.h`

Name	What it does	Header
`kapi_wait_for_exit`	pump the events (sleeping between them) until this window is asked to close	`appkit.h`
`kapi_should_exit`	1 once this window was asked to close (its close box, uk_win_app_toggle), else 0	`appkit.h`
`kapi_draw_text_buf`	Draw kernel-font text into an arbitrary app-mapped 0x00RRGGBB buffer (e.g.	`appkit.h`
`kapi_font_width`	Draw kernel-font text into an arbitrary app-mapped 0x00RRGGBB buffer (e.g.	`appkit.h`
`kapi_font_height`	Draw kernel-font text into an arbitrary app-mapped 0x00RRGGBB buffer (e.g.	`appkit.h`
`kapi_meminfo`	Memory snapshot (KB)	`appkit.h`
`kapi_ram_detail`	ABI v33	`appkit.h`
`kapi_sbrk`	Per-process heap	`appkit.h`
`kapi_list_apps`	The names of the installed apps (the folders SD:apps/.app), one a line, into b (s bytes) -> how many.	`appkit.h`
`kapi_get_datetime`	The local date and time (any pointer may be 0) -> 1 a real date, 0 the clock is not set yet (the time since the boot).	`appkit.h`
`kapi_app_dir`	"SD:apps/<this process's name>.app/" into b (s bytes) -> its length	`appkit.h`
`kapi_write`	b (the first 128 bytes of its n) to the kernel's log as a line of "app" -> n, -1 a bad buffer (fd is not used).	`appkit.h`
`kapi_open`	Open a file to read it (on the card, the RAM volume or a provider's path, relative to the working directory) -> its handle, 0	`appkit.h`

Name	What it does	Header
	failure.	
`kapi_read`	up to n bytes from the file's position (advanced) -> the bytes read (0: the end), -1 error	`appkit.h`
`kapi_fsize`	the file's size in bytes (0xFFFFFFFF: over 4 GB, see kapi_fsize64); 0 for a bad handle	`appkit.h`
`kapi_close`	close a file opened with kapi_open	`appkit.h`
`kapi_save_file`	save_file	`appkit.h`
`kapi_chdir`	Working directory	`appkit.h`
`kapi_getcwd`	Working directory	`appkit.h`
`kapi_opendir`	open a folder to list it -> its handle, 0 failure (not a folder)	`appkit.h`
`kapi_readdir`	the next entry (name, size, is_dir) into *e -> 1, 0 at the end / on error	`appkit.h`
`kapi_closedir`	close a folder opened with kapi_opendir	`appkit.h`
`kapi_mkdir`	mkdir / remove / rename	`appkit.h`
`kapi_remove`	mkdir / remove / rename	`appkit.h`
`kapi_rename`	mkdir / remove / rename	`appkit.h`
`kapi_pipe`	a new pipe (a FIFO in memory) -> its stream handle, 0 failure	`appkit.h`
`kapi_file_in`	a file as a stream to read -> its stream handle, 0 failure	`appkit.h`
`kapi_file_out`	A file as a stream to write, created / emptied or (append != 0) written at its end -> its stream handle, 0 failure.	`appkit.h`
`kapi_stream_read`	Up to n bytes from a stream (a pipe waits for its writer) -> the bytes read, 0 the end / a bad handle.	`appkit.h`

Name	What it does	Header
`kapi_stream_read_nb`	Up to n bytes from a stream without waiting -> > 0 the bytes read, 0 the end / a bad handle, -1 nothing yet.	`appkit.h`
`kapi_stream_write`	n bytes to a stream (a full pipe waits for its reader) -> the bytes written, -1 error.	`appkit.h`
`kapi_stream_close`	close a stream handle (its reference to the stream dropped)	`appkit.h`
`kapi_stream_eof`	tell the stream's readers it has ended (the writer is done); the handle stays open	`appkit.h`
`kapi_proc_done`	1 if the process of kapi_spawn has finished (a bad handle too), 0 if it runs (the handle stays, kapi_wait closes it).	`appkit.h`
`kapi_stdin_read`	up to n bytes from this process's standard input -> the bytes read, 0 the end / none	`appkit.h`
`kapi_stdout_write`	n bytes to this process's standard output (it has none = to the kernel's log, 128 bytes at most) -> the bytes written, -1 error.	`appkit.h`
`kapi_spawn`	Start the program at path with the arguments args, its standard input / output the stream handles in / out (0 = none) -> its process handle (kapi_wait, kapi_pro	`appkit.h`
`kapi_wait`	wait for a spawned process's end -> its exit status (the handle closed), -1 not a process handle	`appkit.h`
`kapi_get_args`	this process's argument string into b (n bytes) ->	`appkit.h`

Name	What it does	Header
	its length	
`kapi_stdin`	This task's own stdin/stdout stream handles (for a shell wiring children).	`appkit.h`
`kapi_stdout`	This task's own stdin/stdout stream handles (for a shell wiring children).	`appkit.h`
`kapi_klog_read`	Read the next kernel log event (real-time tee).	`appkit.h`
`kapi_klog_read_at`	(v100) the same + the event's time since boot in microseconds (*us)	`appkit.h`
`kapi_set_verbose`	Verbose kernel logging	`appkit.h`
`kapi_get_verbose`	Verbose kernel logging	`appkit.h`
`kapi_net_status`	TCP/IP sockets over WLAN (ABI v21).	`appkit.h`
`kapi_tcp_connect`	TCP/IP sockets over WLAN (ABI v21).	`appkit.h`
`kapi_tcp_send`	TCP/IP sockets over WLAN (ABI v21).	`appkit.h`
`kapi_tcp_recv`	TCP/IP sockets over WLAN (ABI v21).	`appkit.h`
`kapi_tcp_close`	TCP/IP sockets over WLAN (ABI v21).	`appkit.h`
`kapi_tcp_listen`	TCP server side (ABI v37).	`appkit.h`
`kapi_tcp_accept`	TCP server side (ABI v37).	`appkit.h`
`kapi_screen_grab`	Remote screen (ABI v38).	`appkit.h`
`kapi_inject_pointer`	Remote screen (ABI v38).	`appkit.h`
`kapi_inject_key`	Remote screen (ABI v38).	`appkit.h`
`kapi_ipc_register`	Named IPC services (ABI v40)	`appkit.h`
`kapi_ipc_lookup`	Named IPC services (ABI v40)	`appkit.h`
`kapi_clipboard_set`	The clipboard replaced by the n bytes of d (64 KB at most, n 0 empties it) of the type CLIP_* -> the bytes kept (0 too for a bad pointer, the clipboard	`appkit.h`

Name	What it does	Header
	then as	
`kapi_clipboard_get`	Up to cap bytes of the clipboard into b, its type and its serial (which changes at every set) -> the content's whole length, 0 empty.	`appkit.h`
`kapi_shutdown`	end the session: the card unmounted, then halt (SHUTDOWN_HALT) or restart; does not return	`appkit.h`
`kapi_fullscreen_begin`	Full-screen apps (ABI v41)	`appkit.h`
`kapi_present_fb`	Full-screen apps (ABI v41)	`appkit.h`
`kapi_fullscreen_end`	Full-screen apps (ABI v41)	`appkit.h`
`kapi_get_modifiers`	The keyboard's modifiers (ABI v42	`appkit.h`
`kapi_inject_modifiers`	The keyboard's modifiers (ABI v42	`appkit.h`
`kapi_net_ping`	Network tools (ABI v43).	`appkit.h`
`kapi_net_resolve`	Network tools (ABI v43).	`appkit.h`
`kapi_net_info`	Network tools (ABI v43).	`appkit.h`
`kapi_vfs_register`	this process serves the paths starting with prefix -> 1, 0 (taken by another, no room)	`appkit.h`
`kapi_vfs_next`	The next request for this provider into *req -> 1, 0 none (with blocking != 0, after waiting up to 0.5 s for one).	`appkit.h`
`kapi_vfs_req_data`	Up to cap bytes of the request id's payload (VFS_OP_SAVE's data) from offset into buf -> the bytes copied, 0 none.	`appkit.h`
`kapi_vfs_reply`	Answer the request id with its status and len bytes of data (0 = none), its caller woken -> 1, 0 (no such request, bad data).	`appkit.h`
`kapi_wlan_scan`	Wi-Fi scan (ABI v45)	`appkit.h`

Name	What it does	Header
`kapi_sound_acquire`	Sound (ABI v46), on the 3.5 mm jack.	`appkit.h`
`kapi_sound_release`	Sound (ABI v46), on the 3.5 mm jack.	`appkit.h`
`kapi_sound_start`	(RETIRED 2026-10-05	`appkit.h`
`kapi_sound_stop`	(RETIRED 2026-10-05	`appkit.h`
`kapi_sound_write`	(RETIRED 2026-10-05	`appkit.h`
`kapi_sound_status`	(RETIRED 2026-10-05	`appkit.h`
`kapi_sound_instrument`	FM (ABI v47)	`appkit.h`
`kapi_key_held`	Held keys (ABI v48), for games (key events only report presses)	`appkit.h`
`kapi_inject_key_held`	Held keys (ABI v48), for games (key events only report presses)	`appkit.h`
`kapi_exec_as`	Run a program under another process name (ABI v49)	`appkit.h`
`kapi_pad_state`	USB gamepads (v50)	`appkit.h`
`kapi_core_acquire`	App cores (v51)	`appkit.h`
`kapi_core_run`	App cores (v51)	`appkit.h`
`kapi_core_state`	App cores (v51)	`appkit.h`
`kapi_core_release`	App cores (v51)	`appkit.h`
`kapi_gpu_info`	The V3D GPU (v52)	`appkit.h`
`kapi_gpu_draw`	The V3D GPU (v52)	`appkit.h`
`kapi_gpu_texture`	The GPU's full pipeline (v53)	`appkit.h`
`kapi_gpu_render`	The GPU's full pipeline (v53)	`appkit.h`
`kapi_fullscreen_direct`	Full screen straight into the displayed framebuffer (v55	`appkit.h`
`kapi_seek`	The read position of an opened file (v57)	`appkit.h`
`kapi_code_alloc`	Writable + executable memory for generated code, a JIT (v58)	`appkit.h`
`kapi_fsize64`	A file's whole size (v59	`appkit.h`
`kapi_vol_info`	(v71) vol_info	`appkit.h`

Name	What it does	Header
`kapi_vol_list`	(v93) The volumes	`appkit.h`
`kapi_vol_eject`	vol_eject	`appkit.h`
`kapi_vol_mount`	vol_mount	`appkit.h`
`kapi_vol_format`	vol_format	`appkit.h`
`kapi_pop_event`	(v73) The event pump's kernel half -- what kapi_pump_events does, step by step, for a pump of the app's own (a protected app's table runs its pump that way, ker	`appkit.h`
`kapi_event_mods`	(v73) The event pump's kernel half -- what kapi_pump_events does, step by step, for a pump of the app's own (a protected app's table runs its pump that way, ker	`appkit.h`
`kapi_pop_post`	(v73) The event pump's kernel half -- what kapi_pump_events does, step by step, for a pump of the app's own (a protected app's table runs its pump that way, ker	`appkit.h`
`kapi_pump_sleep`	(v73) The event pump's kernel half -- what kapi_pump_events does, step by step, for a pump of the app's own (a protected app's table runs its pump that way, ker	`appkit.h`
`kapi_proc_stats`	(v74) A process's system calls (pid 0	`appkit.h`
`kapi_vm_map`	(v75) The POSIX layer's kernel half (docs/POSIX-PLAN.md	`appkit.h`
`kapi_vm_unmap`	(v75) The POSIX layer's kernel half (docs/POSIX-	`appkit.h`

Name	What it does	Header
	PLAN.md	
`kapi_vm_protect`	(v75) The POSIX layer's kernel half (docs/POSIX-PLAN.md	`appkit.h`
`kapi_vm_advise`	(v75) The POSIX layer's kernel half (docs/POSIX-PLAN.md	`appkit.h`
`kapi_vm_query`	(v75) The POSIX layer's kernel half (docs/POSIX-PLAN.md	`appkit.h`
`kapi_vm_stats`	(v75) The POSIX layer's kernel half (docs/POSIX-PLAN.md	`appkit.h`
`kapi_thread_create_ex`	(v75) The POSIX layer's kernel half (docs/POSIX-PLAN.md	`appkit.h`
`kapi_thread_info`	(v75) The POSIX layer's kernel half (docs/POSIX-PLAN.md	`appkit.h`
`kapi_file_open`	v75 WP-FILE/PROC	`appkit.h`
`kapi_file_read`	v75 WP-FILE/PROC	`appkit.h`
`kapi_file_write`	v75 WP-FILE/PROC	`appkit.h`
`kapi_file_seek`	v75 WP-FILE/PROC	`appkit.h`
`kapi_file_truncate`	v75 WP-FILE/PROC	`appkit.h`
`kapi_file_sync`	v75 WP-FILE/PROC	`appkit.h`
`kapi_file_stat`	v75 WP-FILE/PROC	`appkit.h`
`kapi_file_close`	v75 WP-FILE/PROC	`appkit.h`
`kapi_path_stat`	v75 WP-FILE/PROC	`appkit.h`
`kapi_path_unlink`	v75 WP-FILE/PROC	`appkit.h`
`kapi_path_mkdir`	v75 WP-FILE/PROC	`appkit.h`
`kapi_path_rename`	v75 WP-FILE/PROC	`appkit.h`
`kapi_path_utime`	v75 WP-FILE/PROC	`appkit.h`
`kapi_dir_read`	v75 WP-FILE/PROC	`appkit.h`
`kapi_stream_write_nb`	v75 WP-FILE/PROC	`appkit.h`
`kapi_spawn_ex`	v75 WP-FILE/PROC	`appkit.h`
`kapi_proc_wait`	v75 WP-FILE/PROC	`appkit.h`
`kapi_get_argv`	v75 WP-FILE/PROC	`appkit.h`
`kapi_get_env`	v75 WP-FILE/PROC	`appkit.h`
`kapi_getpid`	v75 WP-FILE/PROC	`appkit.h`

Name	What it does	Header
`kapi_clock_info`	v75 WP-FILE/PROC	`appkit.h`
`kapi_sleep_us`	v75 WP-FILE/PROC	`appkit.h`
`kapi_sock_open`	v75 WP-NET	`appkit.h`
`kapi_sock_connect`	v75 WP-NET	`appkit.h`
`kapi_sock_bind`	v75 WP-NET	`appkit.h`
`kapi_sock_listen`	v75 WP-NET	`appkit.h`
`kapi_sock_accept`	v75 WP-NET	`appkit.h`
`kapi_sock_send`	v75 WP-NET	`appkit.h`
`kapi_sock_recv`	v75 WP-NET	`appkit.h`
`kapi_sock_shutdown`	v75 WP-NET	`appkit.h`
`kapi_sock_close`	v75 WP-NET	`appkit.h`
`kapi_sock_getopt`	v75 WP-NET	`appkit.h`
`kapi_sock_setopt`	v75 WP-NET	`appkit.h`
`kapi_sock_name`	v75 WP-NET	`appkit.h`
`kapi_poll`	v75 WP-NET	`appkit.h`
`kapi_sock_pair`	(v76, WP-IPC	`appkit.h`
`kapi_sock_sendmsg`	(v76, WP-IPC	`appkit.h`
`kapi_sock_recvmsg`	(v76, WP-IPC	`appkit.h`
`kapi_shm_create`	(v76, WP-IPC	`appkit.h`
`kapi_shm_open`	(v76, WP-IPC	`appkit.h`
`kapi_shm_unlink`	(v76, WP-IPC	`appkit.h`
`kapi_shm_ctl`	(v76, WP-IPC	`appkit.h`
`kapi_shm_map`	(v76, WP-IPC	`appkit.h`
`kapi_handle_close`	(v76, WP-IPC	`appkit.h`
`kapi_spawn_ex2`	(v76, WP-IPC	`appkit.h`
`kapi_get_handles`	(v76, WP-IPC	`appkit.h`
`kapi_image_preload`	(v77) Program images (docs/02 section 7)	`appkit.h`
`kapi_image_unload`	(v77) Program images (docs/02 section 7)	`appkit.h`
`kapi_image_list`	(v77) Program images (docs/02 section 7)	`appkit.h`
`kapi_kernel_info`	(v79) What the running kernel is	`appkit.h`
`kapi_cpu_stats`	(v80) The cores	`appkit.h`
`kapi_net_stats`	(v80) The bytes pid's sockets received and sent,	`appkit.h`

Name	What it does	Header
	its open sockets (pid 0	
`kapi_lib_open`	(v83) A shared library (docs/SHARED-LIBS-PLAN.md)	`appkit.h`
`kapi_lib_open_as`	(v97) The graphics server's own library under another name (docs/POCKETUI-TECH-STUDY.md section 3)	`appkit.h`
`kapi_sound_output`	(v84) The sound's output	`appkit.h`
`kapi_sound_input`	(v99) The sound's input, a microphone (KAPI_SND_IN_*)	`appkit.h`
`kapi_sound_clients`	(v85) the sound's mixer	`appkit.h`
`kapi_sound_client_volume`	(v85) the sound's mixer	`appkit.h`
`kapi_is_protected`	(v73) Is this process protected (EL0, kern/el0.h)? Its table's memcpy is then user code, next to the table, instead of the kernel's.	`appkit.h`
`kapi_sound_volume`	the master volume 0..10 and mute (-1	`appkit.h`
`kapi_wlan_reconnect`	wpa_supplicant.conf read again + DHCP again, no reboot (-1	`appkit.h`
`kapi_gpu_program`	v61	`appkit.h`
`kapi_gpu_render2`	v61	`appkit.h`
`kapi_gpu_render3`	(v62) gpu_render3	`appkit.h`
`kapi_gpu_vbuf`	(v63) gpu_vbuf	`appkit.h`
`kapi_gpu_texture_rect`	(v70) gpu_texture_rect	`appkit.h`
`kapi_screen_set`	(v66) screen_set	`appkit.h`
`kapi_thread_create`	(v67) Threads	`appkit.h`
`kapi_thread_exit`	(v67) Threads	`appkit.h`
`kapi_thread_join`	(v67) Threads	`appkit.h`
`kapi_thread_self`	(v67) Threads	`appkit.h`
`kapi_mutex_create`	Synchronisation objects (handles > 0	`appkit.h`
`kapi_mutex_lock`	Synchronisation objects (handles > 0	`appkit.h`

Name	What it does	Header
`kapi_mutex_unlock`	Synchronisation objects (handles > 0	`appkit.h`
`kapi_event_create`	Synchronisation objects (handles > 0	`appkit.h`
`kapi_event_set`	Synchronisation objects (handles > 0	`appkit.h`
`kapi_event_reset`	Synchronisation objects (handles > 0	`appkit.h`
`kapi_event_wait`	Synchronisation objects (handles > 0	`appkit.h`
`kapi_barrier_create`	Synchronisation objects (handles > 0	`appkit.h`
`kapi_barrier_wait`	Synchronisation objects (handles > 0	`appkit.h`
`kapi_sync_close`	Synchronisation objects (handles > 0	`appkit.h`
`kapi_post`	Synchronisation objects (handles > 0	`appkit.h`
`kapi_pump_wait`	Synchronisation objects (handles > 0	`appkit.h`
`kapi_lock`	(on an app core -- kapi_core_run's code, which makes no kapi call -- it spins instead)	`appkit.h`
`kapi_unlock`	(on an app core -- kapi_core_run's code, which makes no kapi call -- it spins instead)	`appkit.h`
`kapi_wait_word`	(v68) A futex.	`appkit.h`
`kapi_wake_word`	(v68) A futex.	`appkit.h`
`kapi_thread_priority`	(v68) A futex.	`appkit.h`
`kapi_midi_read`	(v68) USB MIDI input	`appkit.h`
`kapi_midi_devices`	(v68) USB MIDI input	`appkit.h`
`kapi_screen_native`	(v69) kapi_screen_native (&w, &h)	`appkit.h`
`kapi_set_timezone`	(v69) kapi_screen_native (&w, &h)	`appkit.h`
`kapi_clock_us`	The kernel's microsecond clock (CTimer::GetClockTicks	`appkit.h`
`kapi_sound_config`	(v68) Low-latency sound,	`appkit.h`

Name	What it does	Header
	for the sound owner (kapi_sound_acquire).	
`kapi_sound_map`	(v68) Low-latency sound, for the sound owner (kapi_sound_acquire).	`appkit.h`
`kapi_sound_ring_free`	Frames the ring can take now.	`appkit.h`
`kapi_sound_ring_write`	Copy up to n s16 stereo frames into the ring -> the frames taken (no kapi call	`appkit.h`
`kapi_reboot`	Reboot the machine (ABI v25).	`appkit.h`
`kapi_kbd_ready`	1 if a USB keyboard is attached & ready, else 0 (ABI v26).	`appkit.h`
`kapi_set_keymap_data`	Load a keyboard layout from a .kmap blob (ABI v27)	`appkit.h`
`kapi_random`	Hardware RNG (ABI v30)	`appkit.h`
`kapi_surface_create`	Shell surfaces (ABI v35)	`appkit.h`
`kapi_surface_map`	Shell surfaces (ABI v35)	`appkit.h`
`kapi_surface_size`	Shell surfaces (ABI v35)	`appkit.h`
`kapi_surface_present`	Shell surfaces (ABI v35)	`appkit.h`
`kapi_surface_destroy`	Shell surfaces (ABI v35)	`appkit.h`
`kapi_register_shell`	Activity-shell IPC (ABI v35)	`appkit.h`
`kapi_shell_request`	Activity-shell IPC (ABI v35)	`appkit.h`
`kapi_mailbox_send`	Activity-shell IPC (ABI v35)	`appkit.h`
`kapi_mailbox_recv`	Activity-shell IPC (ABI v35)	`appkit.h`
`kapi_ws_ctl`	(v89) The graphics server's own door to the kernel (Elegant, SD:/bin/elegant)	`appkit.h`
`kapi_memset`	Memory primitives (ABI v36)	`appkit.h`
`kapi_memcpy`	Memory primitives (ABI v36)	`appkit.h`
`kapi_memmove`	Memory primitives (ABI v36)	`appkit.h`
`get_ticks`	Friendly aliases used by	`appkit.h`

Name	What it does	Header
	the demos (create_window, present	
`msleep`	Friendly aliases used by the demos (create_window, present	`appkit.h`
`pump_events`	Friendly aliases used by the demos (create_window, present	`appkit.h`
`should_exit`	Friendly aliases used by the demos (create_window, present	`appkit.h`
`ax_strcat`	Strings.	`appkit.h`
`ax_app_path`	Strings.	`appkit.h`
`ax_streq`	Strings.	`appkit.h`
`ax_strlen`	Strings.	`appkit.h`
`ax_itoa`	Strings.	`appkit.h`
`ax_fmt2`	Strings.	`appkit.h`
`ax_puts`	The console	`appkit.h`
`ax_putln`	The console	`appkit.h`
`app_ini_load_path`	load the .ini file at path (replacing the one loaded) -> its number of entries, -1 not opened	`appkit.h`
`app_ini_load`	the same for <the program's own folder>/filename (kapi_app_dir)	`appkit.h`
`app_ini_get`	The value of key in section (0 or "" = the keys before any [section]) of the loaded file, else def.	`appkit.h`
`app_ini_get_int`	The value of key in section as a decimal integer (a sign allowed), def when the key is absent or has no digit.	`appkit.h`
`app_ini_count`	the number of entries (key=value lines) of the loaded file	`appkit.h`
`app_ini_section`	the section of the entry i (0-based, in the file's	`appkit.h`

Name	What it does	Header
	order); "" out of range	
`app_ini_key`	the key of the entry i; "" out of range	`appkit.h`
`app_ini_value`	the value of the entry i; "" out of range	`appkit.h`
`ax_load_keymap`	The keyboard layout ("FR", "BE"...)	`appkit.h`
`lx_low`	c in lower case ('A'..'Z' only)	`appkit.h`
`lx_len`	the length of s (0 for a null pointer)	`appkit.h`
`lx_cat`	s appended to d at *n (advanced), never past cap, NUL-terminated	`appkit.h`
`lx_exists`	1 if the file at path can be opened (kapi_open), else 0	`appkit.h`
`lx_entry`	The i-th "ext = program" line of runners.ini (0-based)	`appkit.h`
`lx_lists_ext`	Does the value of an app.txt's "games" / "opens" (the extensions after each "System:", or all the words of "opens") hold ext?	`appkit.h`
`lx_app_for`	The app (SD:/apps/.app/main) whose app.txt opens files of this extension	`appkit.h`
`lx_runner`	The runner of a file, by its extension (runners.ini, else the built-in list)	`appkit.h`
`lx_cmdline`	The runner's command line	`appkit.h`
`lx_open_as`	A program file	`appkit.h`
`lx_open`	A program file	`appkit.h`
`lx_launch_dir`	An app bundle by folder path (".../x.app")	`appkit.h`
`lx_launch`	An app by name (SD:apps/.app), with arguments (0 / ""	`appkit.h`

`appkit/appkit.h`

appkit.h -- AppKit: what a program includes to talk to the system.

A program calls the functions declared here (kapi_*: the names of AppKit's table, kept as they were); they are AppKit's (SD:/lib/appkit.so, user/Kits/appkit), which the kernel binds to every program. A program never talks to the kernel itself. Their bodies are in appkit_calls.inc, beside this file. The structures and constants shared with the kernel come from <kern/kapi_abi.h>.

(This header was user/kapi.h until 2026-10-05. There is no kapi.h any more: what talks to the kernel is in this folder only.)

how a program reaches the kernel: AppKit (2026-10-05)

The kapi_* functions declared below (KAPI_FN) are AppKit's (SD:/lib/appkit.so, user/Kits/appkit: the ONE interface between the programs and the kernel -- loaded by the kernel and bound to every program with no call of theirs). A program calls them BY NAME, through the import stubs linked into it (lib/appkit_stubs.o). THIS HEADER ONLY DECLARES THEM: their bodies -- the calls through the kernel's table -- are in appkit_calls.inc (beside this file), compiled into AppKit (appkit/appkit.c) and nowhere else. So the kernel's table can be restructured -- entries moved, removed, merged -- by adapting that file and rebuilding AppKit alone: no program changes.

The one exception: with KAPI_INLINE (the tests of the table itself: el0test, faulttest), or in a PC build (the simulator's stand-in kernel has a table of its own and no AppKit), the bodies are compiled inline into the program -- the include at this header's end.

What stays inline here makes no call to the kernel (the spin locks, kapi_clock_us, the sound ring's writer): an app core may use it.

```
#define KAPI_C  extern "C"
#define KAPI_C
#define KAPI_FN static inline
#define KAPI_FN KAPI_C
```

The kernel's kapi version (KAPI_ABI_VERSION as the running kernel says it): what a program tests before a call that came late (it was the first word of the kernel's table).

```
unsigned kapi_abi_version (void);
```

The address of a slot of the kernel's table (its index in 8-byte words): for the tests of the table itself. Nothing else needs it.

```
unsigned long long kapi_table_slot (unsigned slot);
```

Window creation flags (must match kern/gui/window.h).

```
#define WIN_FLAG_BORDERLESS (1u << 0)   // no title bar / border / close box
#define WIN_FLAG_BACKMOST   (1u << 1)   // pinned to the bottom of the z-order (shell
desktop)
#define WIN_FLAG_TOPMOST    (1u << 2)   // pinned to the top, never active (the menu bar)
#define WIN_FLAG_TRANSPARENT    (1u << 3)   // magenta (0xFF00FF) client pixels are see-
through
#define WIN_FLAG_SYSTEM     (1u << 4)   // system component: not listed as an open app (panel
taskbar)
```

```
#define WIN_FLAG_ALPHA      (1u << 5)   // (v64, borderless) the pixels' top byte is a
transparency
```

(0 opaque .. 255 see-through; clicks there go below)

```
#define WIN_FLAG_FIXED      (1u << 6)   // (v69) not movable, no title buttons, kept centred
```

Event kinds (must match kern/gui/window.h).

```
#define GUI_EVENT_CLICK     1
#define GUI_EVENT_CHECK_CHANGED 2
#define GUI_EVENT_TEXT_CHANGED  3
#define GUI_EVENT_VALUE_CHANGED 4
#define GUI_EVENT_KEY       5   // key pressed; value = char or KEY_* code
#define GUI_EVENT_CANVAS_CLICK  6   // client-area press; value = (buttons<<32)|(x<<16)|y
#define GUI_EVENT_CANVAS_MOTION 7   // drag (button held) over the client area; same value
```

buttons: bit0 left, bit1 right Full pointer stream (ABI v22, opt-in via uk_win_on_pointer) for app-side widget toolkits (uikit.h). value packs (wheel<<48)|(changed<<40)|(buttons<<32)|(x<<16)|y, all client-relative; decode with the GUI_PTR_* macros below.

```
#define GUI_EVENT_PTR_MOVE  8   // cursor moved
#define GUI_EVENT_PTR_DOWN  9   // a button went down (changed = which: 1/2/4)
#define GUI_EVENT_PTR_UP    10  // a button went up (changed = which)
#define GUI_EVENT_PTR_ENTER 11  // cursor entered the client area
#define GUI_EVENT_PTR_LEAVE 12  // cursor left the client area
#define GUI_EVENT_PTR_WHEEL 13  // scroll wheel turned (GUI_PTR_WHEEL = signed notch delta)
#define GUI_EVENT_MENU      14  // menu-bar command chosen (value = item id, ABI v39)
#define MENU_QUIT       (-1)    // uk_win_menu_command id: close the active app
```

Drag & drop (ABI v42) -- to the pointer handler (see uk_win_drag_begin):

```
#define GUI_EVENT_DROP      15  // dropped on us: GUI_PTR_X/Y + GUI_DND_FLAGS;
uk_win_drag_data
#define GUI_EVENT_DRAG_OVER 16  // a drag hovers us (GUI_DND_FLAGS & DND_F_LEAVE: it left)
#define GUI_EVENT_DRAG_DONE 17  // to the source: GUI_DND_PID (0 = none) + GUI_DND_FLAGS
#define GUI_EVENT_DISPLAY_RESIZE 19 // (v66) the screen's size changed: GUI_DISPLAY_W / _H
(value)
#define GUI_DISPLAY_W(v)    ((int) (((v) >> 16) & 0xFFFF))
#define GUI_DISPLAY_H(v)    ((int) ((v) & 0xFFFF))
#define GUI_EVENT_WINCTL    18  // (v64) a title button: value KAPI_FRAME_MENU (the window
menu)
```

or KAPI_FRAME_MAXIMISE (also a double click on the title)

```
#define GUI_DND_FLAGS(v)    ((unsigned) (((unsigned long long) (v) >> 32) & 0xFF))
#define GUI_DND_PID(v)      ((int) ((unsigned long long) (v) & 0xFFFFFFFF))
#define DND_F_COPY      1   // Ctrl held: copy instead of move
#define DND_F_CANCEL        2   // DRAG_DONE: cancelled (Esc)
#define DND_F_DESKTOP       4   // DRAG_DONE: dropped on the desktop / no window
#define DND_F_LEAVE     8   // DRAG_OVER: the drag left this window
#define DND_TEXT        1   // payload types (= CLIP_TEXT / CLIP_FILES)
#define DND_FILES       2
#define MOD_CTRL        1   // kapi_get_modifiers
#define MOD_SHIFT       2
#define MOD_ALT         4
#define WIN_MENU_MAX_USER   2048    // max menu spec length (= the kernel's WIN_MENU_MAX)
#define GUI_PTR_Y(v)        ((int) ((unsigned long long) (v) & 0xFFFF))
#define GUI_PTR_X(v)        ((int) (((unsigned long long) (v) >> 16) & 0xFFFF))
#define GUI_PTR_BUTTONS(v)  ((int) (((unsigned long long) (v) >> 32) & 0xFF))   // held mask
```

```
#define GUI_PTR_CHANGED(v)  ((int) (((unsigned long long) (v) >> 40) & 0xFF))   // 1 left/2 right/4 mid
#define GUI_PTR_WHEEL(v)    ((int) (signed char) (((unsigned long long) (v) >> 48) & 0xFF))  // +fwd/-back
```

Logical key codes (GUI_EVENT_KEY value). Printable keys are their ASCII value.

```
#define KEY_BACKSPACE       8
#define KEY_TAB         9
#define KEY_ENTER       13
#define KEY_UP          0x100
#define KEY_DOWN        0x101
#define KEY_LEFT        0x102
#define KEY_RIGHT       0x103
#define KEY_HOME        0x104
#define KEY_END         0x105
#define KEY_PGUP        0x106
#define KEY_PGDN        0x107
#define KEY_DEL         0x108
#define KEY_F1          0x110   // .. KEY_F12 = 0x11B (KEY_F1 + n - 1)
#define KEY_F12         0x11B
#define KEY_PRTSC       0x11C   // (2026-10-26) Print Screen and Pause: HELD keys only (kapi_key_held), no key event
#define KEY_PAUSE       0x11D   //   -- the PocketTerm35's Select and Start, mapped in the Gamepad applet
```

windowing

The windows are UIKit's (2026-10-08): uk_win_create, uk_win_present, uk_win_menu_set, uk_win_list... in uikit/win.h -- the graphics server (Elegant; PocketUI to come) is spoken to by each server's UIKit, its port. The flags, the events and the keys above and the structures of kern/kapi_abi.h keep their place here: the events come through the kernel's pump (kapi_pump_events, kapi_pump_wait, kapi_should_exit).

```
int kapi_launch (const char *n);    // start the app n (SD:apps/<n>.app/main) as a new process -> 1, 0 failure
```

Every task, one line each " " (state R / S / B / N, kind a = an app or k = a kernel task), into b (s bytes) -> how many.

```
int kapi_list_tasks (char *b, unsigned s);
int kapi_kill (const char *name);   // kill the app of that name -> 1, 0 (not running, a kernel task, the caller)
```

ps / kill by PID. list_procs: lines " <a|k>   " (pages: its 64 KB frames). kill_pid: force 0 = clean close, 1 = hard terminate; 1 ok / 0 no such pid / -1 protected.

```
int kapi_list_procs (char *b, unsigned s);
int kapi_kill_pid (int pid, int force);
```

(v91) A process's tree: its descendants (the processes it spawned, the ones they spawned...). op KAPI_TREE_LIST -> how many descendants pid has (up to cap of their pids into out); KAPI_TREE_KILL: pid and all of them terminated now, the leaves first -> how many were; KAPI_TREE_KILL_CHILDREN: its descendants only. -KAPI_ESRCH no such process, -KAPI_EPERM a kill that would take the caller, -KAPI_EINVAL, -KAPI_ENOSYS (a kernel before v91). A terminal closing a tab: its shell and all it runs.

```
int kapi_proc_tree (int pid, int op, int *out, unsigned cap);
```

Keyboard layout: switch among the compiled-in country maps; read the current one.

```
int kapi_set_keymap (const char *name);
int kapi_get_keymap (char *b, unsigned s);
```

Modal-dialog button sets (used by the user-side ui::MessageBox in uidialog.hpp; the kernel message_box/file dialogs were removed -- the toolkit draws its own).

```
#define MB_OK       0
#define MB_OKCANCEL 1
#define MB_YESNO    2
#define MB_YESNOCANCEL  3   // uikit: Yes = 1, No = 2, Cancel / Esc = 0
```

Run an ELF at an absolute path with an argv string (fire-and-forget). 1/0.

```
int kapi_exec (const char *path, const char *args);
```

Framebuffer size in pixels (for edge-pinned borderless windows).

```
void kapi_screen_size (int *w, int *h);
unsigned kapi_get_ticks (void);
void kapi_msleep (unsigned ms);
void kapi_yield (void);
void kapi_exit (int s);
```

(The kernel-drawn widget wrappers -- kapi_add*_/kapi_widget_* -- were removed: every app now builds its UI with the user-side uikit.hpp toolkit. See uikit.hpp.)

events

Run what is pending -- the kapi_post calls, then this program's events through their handlers -- and return (no wait).

```
void kapi_pump_events (void);
void kapi_wait_for_exit (void); // pump the events (sleeping between them) until this window
is asked to close
int kapi_should_exit (void);    // 1 once this window was asked to close (its close box,
uk_win_app_toggle), else 0
```

kernel-font text

Draw kernel-font text into an arbitrary app-mapped 0x00RRGGBB buffer (e.g. a window- chrome copy from uk_win_chrome). Transparent background; dst must be a user VA.

```
void kapi_draw_text_buf (unsigned *dst, int dw, int dh, int x, int y, const char *s, unsigned
c);
int kapi_font_width (void);
int kapi_font_height (void);
```

Memory snapshot (KB): total RAM, free, app-owned, page size. Any pointer may be 0.

```
int kapi_meminfo (unsigned long *total_kb, unsigned long *free_kb, unsigned long *app_kb,
unsigned *page_kb);
```

ABI v33: firmware-detected board RAM + app page-pool (HIGH zone) total/free, the bytes reclaimed above 4GB, and the high-segment count. All KB; any pointer may be 0. (detected = physical board RAM e.g. 8192 MB; apppool = the zone backing app frames via palloc_high.)

```
int kapi_ram_detail (unsigned long *detected_kb, unsigned long *apppool_kb, unsigned long
*apppool_free_kb, unsigned long *above4g_kb, unsigned *nsegments);
```

Per-process heap: move the break by `inc` bytes (Unix sbrk); returns the previous break or (void*)-1. The user allocator (umm.h) is built on this; apps rarely call it.

```
void * kapi_sbrk (long inc);
```

enumeration + clock

The names of the installed apps (the folders SD:apps/.app), one a line, into b (s bytes) -> how many.

```
int kapi_list_apps (char *b, unsigned s);
```

The local date and time (any pointer may be 0) -> 1 a real date, 0 the clock is not set yet (the time since the boot).

```
int kapi_get_datetime (int *y, int *mo, int *d, int *h, int *mi, int *se);
int kapi_app_dir (char *b, unsigned s); // "SD:apps/<this process's name>.app/" into b (s bytes) -> its length
```

console + files

b (the first 128 bytes of its n) to the kernel's log as a line of "app" -> n, -1 a bad buffer (fd is not used).

```
int kapi_write (int fd, const void *b, unsigned n);
```

Open a file to read it (on the card, the RAM volume or a provider's path, relative to the working directory) -> its handle, 0 failure.

```
void * kapi_open (const char *p);
int kapi_read (void *h, void *b, unsigned n);   // up to n bytes from the file's position (advanced) -> the bytes read (0: the end), -1 error
unsigned kapi_fsize (void *h);  // the file's size in bytes (0xFFFFFFFF: over 4 GB, see kapi_fsize64); 0 for a bad handle
void kapi_close (void *h);  // close a file opened with kapi_open
```

save_file: the whole file written (created / replaced): the bytes written (>= 0; an empty file: 0 -- test < 0 for a failure, or == n), or -1.

```
int kapi_save_file (const char *p, const void *b, unsigned n);
```

Working directory: chdir (1/0) + getcwd. Relative paths in file ops resolve here.

```
int kapi_chdir (const char *p);
int kapi_getcwd (char *b, unsigned n);
```

directory listing

```
void * kapi_opendir (const char *p);    // open a folder to list it -> its handle, 0 failure (not a folder)
int kapi_readdir (void *d, struct kapi_dirent *e);  // the next entry (name, size, is_dir) into *e -> 1, 0 at the end / on error
void kapi_closedir (void *d);   // close a folder opened with kapi_opendir
```

mkdir / remove / rename: 0 = success, -1 = failure (FatFs result).

```
int kapi_mkdir (const char *p);
int kapi_remove (const char *p);
int kapi_rename (const char *from, const char *to);
```

stdio / streams / processes

```
void * kapi_pipe (void);    // a new pipe (a FIFO in memory) -> its stream handle, 0 failure
void * kapi_file_in (const char *p);    // a file as a stream to read -> its stream handle, 0
failure
```

A file as a stream to write, created / emptied or (append != 0) written at its end -> its stream handle, 0 failure.

```
void * kapi_file_out (const char *p, int append);
```

Up to n bytes from a stream (a pipe waits for its writer) -> the bytes read, 0 the end / a bad handle.

```
int kapi_stream_read (void *h, void *b, unsigned n);
```

Up to n bytes from a stream without waiting -> > 0 the bytes read, 0 the end / a bad handle, -1 nothing yet.

```
int kapi_stream_read_nb (void *h, void *b, unsigned n);
```

n bytes to a stream (a full pipe waits for its reader) -> the bytes written, -1 error.

```
int kapi_stream_write (void *h, const void *b, unsigned n);
void kapi_stream_close (void *h);   // close a stream handle (its reference to the stream
dropped)
void kapi_stream_eof (void *h); // tell the stream's readers it has ended (the writer is
done); the handle stays open
```

1 if the process of kapi_spawn has finished (a bad handle too), 0 if it runs (the handle stays, kapi_wait closes it).

```
int kapi_proc_done (void *proc);
int kapi_stdin_read (void *b, unsigned n);  // up to n bytes from this process's standard
input -> the bytes read, 0 the end / none
```

n bytes to this process's standard output (it has none = to the kernel's log, 128 bytes at most) -> the bytes written, -1 error.

```
int kapi_stdout_write (const void *b, unsigned n);
```

Start the program at path with the arguments args, its standard input / output the stream handles in / out (0 = none) -> its process handle (kapi_wait, kapi_proc_done), 0 failure.

```
void * kapi_spawn (const char *path, const char *args, void *in, void *out);
int kapi_wait (void *proc); // wait for a spawned process's end -> its exit status (the
handle closed), -1 not a process handle
int kapi_get_args (char *b, unsigned n);    // this process's argument string into b (n
bytes) -> its length
```

This task's own stdin/stdout stream handles (for a shell wiring children). 0 = none.

```
void * kapi_stdin (void);
void * kapi_stdout (void);
```

Read the next kernel log event (real-time tee). 1 + fills fields, or 0 if empty.

```
int kapi_klog_read (int *sev, char *src, unsigned sc, char *msg, unsigned mc);
```

(v100) the same + the event's time since boot in microseconds (*us); -KAPI_ENOSYS on a kernel before v100.

```
int kapi_klog_read_at (int *sev, unsigned long long *us, char *src, unsigned sc, char *msg,
unsigned mc);
```

Verbose kernel logging: toggle (1/0) at runtime + read the current state.

```
int kapi_set_verbose (int on);
int kapi_get_verbose (void);
```

TCP/IP sockets over WLAN (ABI v21). net_status: 1 + dotted IP into ip[] if the link is up, else 0. tcp_connect: host = dotted-quad or DNS name; >=0 handle / <0 error. tcp_send: blocking, the bytes queued / <0 -- a short count: a 5 s timeout after those (they are sent; send the rest again). tcp_recv: NON-BLOCKING -- >0 bytes, 0 nothing yet, <0 closed/error. tcp_close: drop the connection.

```
int kapi_net_status (char *ip, unsigned cap);
int kapi_tcp_connect (const char *host, unsigned port);
int kapi_tcp_send (int sock, const void *buf, unsigned len);
int kapi_tcp_recv (int sock, void *buf, unsigned len);
void kapi_tcp_close (int sock);
```

TCP server side (ABI v37). tcp_listen: listening handle >=0 / <0 (-6 port in use). tcp_accept: BLOCKS until a peer connects; connected handle >=0 + peer IP / <0.

```
int kapi_tcp_listen (unsigned port);
int kapi_tcp_accept (int listen_sock, char *ip, unsigned cap);
```

Remote screen (ABI v38). screen_grab: composite the screen into dst (w*h 0x00RRGGBB, w/h = kapi_screen_size) -> 1 / 0, or 2 = unchanged since the previous grab into the same dst (left as is: skip the diff). inject_pointer/inject_key: input as if from the USB mouse (buttons bit0 L, bit1 R, bit2 M; wheel = notches) / keyboard (cooked string).

```
int kapi_screen_grab (unsigned *dst, int w, int h);
void kapi_inject_pointer (int x, int y, unsigned buttons, int wheel);
void kapi_inject_key (const char *keys);
```

Named IPC services (ABI v40): ipc_register -> become service `name` (1 / 0 taken); ipc_lookup -> its pid or 0. Messages go through kapi_mailbox_send / _recv (<= 512 B).

```
int kapi_ipc_register (const char *name);
int kapi_ipc_lookup (const char *name);
```

System clipboard (ABI v40): see clipboard.h for the helpers. Types:

```
#define CLIP_TEXT   1       // plain text
#define CLIP_FILES  2       // file/folder paths, '\n'-separated (copied)
#define CLIP_FILES_CUT  3       // same, cut (the paste moves them)
```

The clipboard replaced by the n bytes of d (64 KB at most, n 0 empties it) of the type CLIP_* -> the bytes kept (0 too for a bad pointer, the clipboard then as it was).

```
int kapi_clipboard_set (int type, const void *d, unsigned n);
```

Up to cap bytes of the clipboard into b, its type and its serial (which changes at every set) -> the content's whole length, 0 empty. b, type and serial may each be 0.

```
int kapi_clipboard_get (int *type, void *b, unsigned cap, unsigned *serial);
```

End of session (0 = halt, 1 = restart).

```
#define SHUTDOWN_HALT       0
#define SHUTDOWN_RESTART    1
void kapi_shutdown (int mode);  // end the session: the card unmounted, then halt
(SHUTDOWN_HALT) or restart; does not return
```

Full-screen apps (ABI v41): fullscreen_begin -> a screen-sized 0x00RRGGBB back buffer (w/h filled); the desktop stops drawing and all input (screen coords) comes to you. Draw, present_fb to show it; fullscreen_end (or exiting) gives the desktop back. (2026-10-08) The kernel's primitives: a program calls UIKit's uk_win_fullscreen_begin / _end (uikit/win.h), which tells its graphics server first, then calls these. kapi_fullscreen_begin no longer tells the server.

```
unsigned * kapi_fullscreen_begin (int *w, int *h);
void kapi_present_fb (void);
void kapi_fullscreen_end (void);
```

The keyboard's modifiers (ABI v42; drag and drop is UIKit's: uk_win_drag_begin / *data). get_modifiers: MOD** held now; inject_modifiers: set them (vncd).

```
unsigned kapi_get_modifiers (void);
void kapi_inject_modifiers (unsigned mods);
```

Network tools (ABI v43). net_ping: one ICMP echo (RTT in us, or -1 down / -3 unresolved / -4 timeout / -5 send failed). net_resolve: DNS -> dotted IP (1/0). net_info: netstat text.

```
int kapi_net_ping (const char *host, unsigned seq, unsigned timeout_ms, char *ip, unsigned
cap);
int kapi_net_resolve (const char *host, char *ip, unsigned cap);
int kapi_net_info (char *buf, unsigned cap);
```

User-space file-system providers (ABI v44, kern/vfs.h): a provider app serves every path starting with its prefix ("FTP:"); the file kapis on those paths reach it as requests.

```
#define VFS_OP_OPEN 1   // path -> status = fid (>= 0); data = u32 file size
#define VFS_OP_READ 2   // a0 fid, a1 offset, a2 length -> data, status = n
#define VFS_OP_CLOSE    3   // a0 fid
#define VFS_OP_LIST 4   // path -> data = (u32 size LE, u8 is_dir, name '\0')*, status =
count
#define VFS_OP_SAVE 5   // path, payload (vfs_req_data) -> status = bytes written
#define VFS_OP_MKDIR    6   // path -> 0 / -1
#define VFS_OP_REMOVE   7   // path -> 0 / -1
#define VFS_OP_RENAME   8   // path -> path2: 0 / -1
int kapi_vfs_register (const char *prefix); // this process serves the paths starting with
prefix -> 1, 0 (taken by another, no room)
```

The next request for this provider into *req -> 1, 0 none (with blocking != 0, after waiting up to 0.5 s for one).

```
int kapi_vfs_next (struct kapi_vfs_req *req, int blocking);
```

Up to cap bytes of the request id's payload (VFS_OP_SAVE's data) from offset into buf -> the bytes copied, 0 none.

```
int kapi_vfs_req_data (unsigned id, void *buf, unsigned cap, unsigned offset);
```

Answer the request id with its status and len bytes of data (0 = none), its caller woken -> 1, 0 (no such request, bad data).

```
int kapi_vfs_reply (unsigned id, int status, const void *data, unsigned len);
```

Wi-Fi scan (ABI v45): the access points around, strongest first (struct kapi_wlan_ap: ssid, bssid, security WLAN_SEC_OPEN/WEP/WPA/WPA2, channel, freq MHz, level dBm, connected). Takes ~3 s (blocks the caller); returns how many (0 = none / no Wi-Fi).

```
int kapi_wlan_scan (struct kapi_wlan_ap *out, int max);
```

Sound (ABI v46), on the 3.5 mm jack. First acquire the output (1 ok, 0 another app has it, -1 no audio); only the owner plays, and its release -- or its exit -- silences it.

```
  kapi_sound_start (voice 0..15, milli-Hz (440 Hz = 440000), SOUND_SQUARE/SINE/TRIANGLE/
                    SAW/NOISE, volume 0..255): the note plays until kapi_sound_stop (voice)
                    (-1 = all). Short attack / release ramps: no clicks.
  kapi_sound_write (s16 L/R frames at SOUND_RATE, n): a PCM stream mixed with the voices;
                    returns the frames taken (0 = full, retry later) -- audio players.
```

```
int kapi_sound_acquire (void);
void kapi_sound_release (void);
```

(RETIRED 2026-10-05: the voices are AudioKit's -- ak_fm_start / ak_fm_stop / ak_fm_instrument, audiokit/audiokit.h --; these three answer -1)

```
int kapi_sound_start (int voice, unsigned millihz, int wave, int volume);
int kapi_sound_stop (int voice);
int kapi_sound_write (const short *frames, unsigned n);
int kapi_sound_status (unsigned *rate, unsigned *free_frames, unsigned *owner);
```

FM (ABI v47): give a voice a 2-operator FM instrument (struct kapi_fm_instrument, see kern/kapi_abi.h), then kapi_sound_start (voice, milliHz, SOUND_FM, volume) keys it on and kapi_sound_stop (voice) keys it off (its release rate fades it out).

```
int kapi_sound_instrument (int voice, const struct kapi_fm_instrument *ins);
```

Held keys (ABI v48), for games (key events only report presses): 1 while `key` is held and this window has the keyboard. key = KEY_UP/DOWN/LEFT/RIGHT, KEY_ENTER, 27 (Esc), ' ', 'a'..'z' (US position of the key on a USB keyboard), '0'..'9'; since 2026-10-26 also KEY_BACKSPACE, KEY_TAB, KEY_HOME/END/PGUP/PGDN/DEL, KEY_F1..KEY_F12, KEY_PRTSC, KEY_PAUSE. Returns 0 on an older kernel.

```
int kapi_key_held (int key);
void kapi_inject_key_held (int key, int down);
```

Run a program under another process name (ABI v49): a runner running an app is named after the app (its window, list_windows, raise_app). See launch.h.

```
int kapi_exec_as (const char *path, const char *args, const char *name);
```

USB gamepads (v50): the raw state of pad 0..3 (1 = there); user/Include/gamepad.h maps its buttons.

```
int kapi_pad_state (int index, struct kapi_pad *out);
```

App cores (v51): acquire core 2 or 3, run a function of this app there (no kapi call and no malloc in it: compute, and exchange data through memory), poll its state, release.

```
int kapi_core_acquire (void);
int kapi_core_run (int core, void (*fn) (void *), void *arg, void *stack_top);
```

```
int kapi_core_state (int core);
void kapi_core_release (int core);
```

The V3D GPU (v52): gpu_info brings it up (1 = usable, buf says what / why not); gpu_draw renders a depth-tested triangle list (NDC positions + RGBA8 colours) into pixels.

```
int kapi_gpu_info (char *buf, unsigned cap);
int kapi_gpu_draw (const struct kapi_gpu_vertex *v, unsigned n, unsigned clear, unsigned
*pixels, int w, int h, int stride);
```

The GPU's full pipeline (v53): textures (0xAARRGGBB; handle < 0 = a new one, pixels 0 = free), and gpu_render: batches (each: its vertices, matrix, texture, blending, depth test, culling) drawn in one frame into f->pixels. See kern/kapi_abi.h.

```
int kapi_gpu_texture (int handle, const unsigned *pixels, int w, int h, int stride);
int kapi_gpu_render (const struct kapi_gpu_frame *f, const struct kapi_gpu_vertex3 *v,
unsigned nv,
                     const struct kapi_gpu_batch *b, unsigned nb);
```

Full screen straight into the displayed framebuffer (v55; after kapi_fullscreen_begin): its pixels (stride in pixels), or 0 (keep drawing into the back buffer + present_fb).

```
unsigned * kapi_fullscreen_direct (int *w, int *h, int *stride);
```

The read position of an opened file (v57): 0, or -1 (an older kernel, a file not seekable).

```
int kapi_seek (void *h, unsigned long long pos);
```

Writable + executable memory for generated code, a JIT (v58): 0 on an older kernel / full. After writing code: clean the D-cache / invalidate the I-cache over it (__builtin___clear_cache).

```
void * kapi_code_alloc (unsigned long size);
```

A file's whole size (v59: over 4 GB on exFAT; an older kernel: fsize).

```
unsigned long long kapi_fsize64 (void *h);
```

(v71) vol_info: a volume's room -- "SD:", "SD1:", "RAM:" (the RAM volume: files in memory until the Pi restarts), a path on it -> 0 and *out (total / free / used bytes, its type, KAPI_VOL_RAM), -1 no such volume / an older kernel. (RAM: paths work with every file call above from v71.)

```
int kapi_vol_info (const char *path, struct kapi_vol_info *out);
```

(v93) The volumes: SD:, SD1:..SD3: (the card's partitions), USB1:, USB2:, USB3: (USB sticks and disks whole; USB1P1:, USB1P2:... the partitions of one that has several; USB: is USB1:; mounted when plugged in), RAM:. vol_list: every volume (struct kapi_volume: its state KAPI_VST_, *flags KAPI_VF_*, sizes, type, label, the files open on it; flags KAPI_VOLS_ROOM: the free space too) -> how many there are (out: up to max). A USB volume pulled out stays listed as KAPI_VST_REMOVED (KAPI_VF_UNSAFE if it was mounted) until a device takes its place; its gen changes at each event.

```
int kapi_vol_list (struct kapi_volume *out, int max, unsigned flags);
```

vol_eject: a USB device ("USB1:", or any of its partitions: all of them) made safe to remove -- its written files synced, the device's cache flushed, unmounted -> 0; -KAPI_EBUSY files are open on it (synced, still mounted; KAPI_EJECT_FORCE: ejected anyway, they then fail with -KAPI_EIO), -KAPI_EINVAL not a removable volume, -KAPI_ENOENT not mounted.

```
int kapi_vol_eject (const char *vol, unsigned flags);
```

vol_mount: a USB volume ejected but still plugged in (or SD1..SD3) mounted again -> 0 / -KAPI_E*.

```
int kapi_vol_mount (const char *vol);
```

vol_format: the volume emptied, a new file system made (struct kapi_format: KAPI_FMT_AUTO / FAT / FAT32 / EXFAT, the cluster, the label) and mounted -> 0, -KAPI_EPERM (SD:, never; SD1..SD3 without KAPI_FMT_CARD), -KAPI_EBUSY (files open: KAPI_FMT_FORCE), -KAPI_ENODEV, -KAPI_EINVAL (the label: 11 characters, none of "*+,./:;<=>?[]|), -KAPI_ENOSPC (too small or too big for it), -KAPI_EIO. The caller waits while it runs (seconds on a big stick).

```
int kapi_vol_format (const char *vol, const struct kapi_format *fmt);
```

(v73) The event pump's kernel half -- what kapi_pump_events does, step by step, for a pump of the app's own (a protected app's table runs its pump that way, kern/el0.h). pop_event: the window's next event -> 1 (*ev; its handler NOT called), 0 none; event_mods: what kapi_get_modifiers says while a key handler runs (ev->mods), returns the previous value to put back; pop_post: the next kapi_post call -> 1 (*p, not run), 0 none; pump_sleep: kapi_pump_wait without the pump (-> how many are pending). An older kernel: 0 / 0xFFFFFFFF / 0 / 0 (no sleep).

```
int kapi_pop_event (struct kapi_event *ev);
unsigned kapi_event_mods (unsigned mods);
int kapi_pop_post (struct kapi_posted *p);
int kapi_pump_sleep (unsigned timeout_ms);
```

(v74) A process's system calls (pid 0: the caller): the total, the rate per second, the 8 table slots most called (user/BinUtils/kapi_names.h names them) -> 0, -1 no such process / older kernel, -2 bad pointer.

```
int kapi_proc_stats (int pid, struct kapi_syscall_stats *out);
```

(v75) The POSIX layer's kernel half (docs/POSIX-PLAN.md; the structures and KAPI_* values in kern/kapi_abi.h). Every call returns >= 0 on success, -KAPI_Exxx (newlib's errno value) on failure, and -KAPI_ENOSYS on an older kernel or until the call is implemented. libonyxposix (user/Runtime/libc/posix) wraps them as the POSIX functions. v75 WP-MEM: virtual memory (vm_map = mmap, vm_protect = mprotect, vm_advise = madvise...), threads with their stack size and TLS (struct kapi_thread_attr).

```
long long kapi_vm_map (unsigned long long addr, unsigned long long len, unsigned prot,
unsigned flags);
int kapi_vm_unmap (unsigned long long addr, unsigned long long len);
int kapi_vm_protect (unsigned long long addr, unsigned long long len, unsigned prot);
int kapi_vm_advise (unsigned long long addr, unsigned long long len, int advice);
int kapi_vm_query (unsigned long long addr, struct kapi_vm_region *out);
int kapi_vm_stats (int pid, struct kapi_vm_stats *out);
int kapi_thread_create_ex (const struct kapi_thread_attr *attr);
int kapi_thread_info (int tid, struct kapi_thread_info *out);
```

v75 WP-FILE/PROC: open files with 64-bit offsets (pread / pwrite), stat, unlink / rename of open files, dir_read (255-character names), non-blocking pipe writes; spawn_ex / proc_wait, argv / environment blocks, getpid, clock_info (CNTPCT + UTC sample), sleep_us. file_read / file_write: off -1 = at the handle's offset (advanced; O_APPEND writes at the end), else a pread / pwrite; a write past the end fills the gap with zeros. A file unlinked or renamed while open stays usable through its handles. proc_wait (h, KAPI_WAIT_NOHANG, &st) on a running child -> 0 with st.pid set and st.reason -1 (spawn_ex returns once the child has its pid). get_argv / get_env (0, 0) -> the block's size. docs/02 §8 "v75: files and processes".

```
long long kapi_file_open (const char *path, unsigned flags, unsigned mode);
long long kapi_file_read (long long h, void *buf, unsigned long long len, long long off);
long long kapi_file_write (long long h, const void *buf, unsigned long long len, long long
off);
long long kapi_file_seek (long long h, long long off, int whence);
int kapi_file_truncate (long long h, long long size);
int kapi_file_sync (long long h);
int kapi_file_stat (long long h, struct kapi_stat *out);
int kapi_file_close (long long h);
int kapi_path_stat (const char *path, struct kapi_stat *out);
int kapi_path_unlink (const char *path, unsigned flags);
int kapi_path_mkdir (const char *path, unsigned mode);
int kapi_path_rename (const char *from, const char *to);
int kapi_path_utime (const char *path, long long mtime);
int kapi_dir_read (void *dir, struct kapi_dirent2 *out);
int kapi_stream_write_nb (void *h, const void *buf, unsigned len);
long long kapi_spawn_ex (const struct kapi_spawn_attr *a);
int kapi_proc_wait (void *proc, unsigned flags, struct kapi_proc_status *out);
int kapi_get_argv (char *buf, unsigned cap);
int kapi_get_env (char *buf, unsigned cap);
int kapi_getpid (int which);
int kapi_clock_info (struct kapi_clock_info *out);
int kapi_sleep_us (unsigned long long us);
```

v75 WP-NET: BSD sockets (IPv4 TCP / UDP: non-blocking connect, accept, MSG_PEEK...) and poll over sockets, streams and files.

```
int kapi_sock_open (int type, unsigned flags);
int kapi_sock_connect (int s, const struct kapi_sockaddr *to);
int kapi_sock_bind (int s, const struct kapi_sockaddr *addr);
int kapi_sock_listen (int s, int backlog);
int kapi_sock_accept (int s, struct kapi_sockaddr *peer, unsigned flags);
long long kapi_sock_send (int s, const void *buf, unsigned long long len, unsigned flags,
const struct kapi_sockaddr *to);
long long kapi_sock_recv (int s, void *buf, unsigned long long len, unsigned flags, struct
kapi_sockaddr *from);
int kapi_sock_shutdown (int s, int how);
int kapi_sock_close (int s);
int kapi_sock_getopt (int s, int opt, int *value);
int kapi_sock_setopt (int s, int opt, int value);
int kapi_sock_name (int s, int peer, struct kapi_sockaddr *out);
int kapi_poll (struct kapi_pollfd *fds, unsigned n, int timeout_ms);
```

(v76, WP-IPC: docs/POSIX-PLAN.md §14) Local sockets (sock_pair: STREAM / SEQPACKET / DGRAM; numbers >= KAPI_SOCK_LOCAL_BASE, served by every sock_* call and poll), messages carrying handles (sock_sendmsg / sock_recvmsg: struct kapi_msghdr, struct kapi_handle_xfer), shared memory objects (shm_create / shm_open / shm_unlink / shm_ctl / shm_map), handle_close, spawn_ex2 (handles given to the child) and get_handles (the child's side). -KAPI_ENOSYS on an older kernel.

```
int kapi_sock_pair (int type, unsigned flags, int *sv);
long long kapi_sock_sendmsg (int s, const struct kapi_msghdr *m, unsigned flags);
long long kapi_sock_recvmsg (int s, struct kapi_msghdr *m, unsigned flags);
long long kapi_shm_create (unsigned long long size, unsigned flags);
long long kapi_shm_open (const char *name, unsigned oflags, unsigned mode);
int kapi_shm_unlink (const char *name);
long long kapi_shm_ctl (long long h, int op, unsigned long long arg);
long long kapi_shm_map (long long h, unsigned long long addr, unsigned long long len,
unsigned prot,
                        unsigned flags, unsigned long long off);
```

```
int kapi_handle_close (long long h);
long long kapi_spawn_ex2 (const struct kapi_spawn_attr *a, const struct kapi_handle_xfer
*handles, unsigned n);
int kapi_get_handles (struct kapi_handle_xfer *out, unsigned cap);
```

(v77) Program images (docs/02 section 7): a program is loaded once and shared by its processes; its key is its canonical path (lower case, the volume first). image_preload: loaded ahead and kept (returns at once; a run of that path then reads nothing from the card); image_unload: its pin and its name taken away (freed with its last process); image_list: the live images (path 0), or the one a run of path would map (1 / 0). -KAPI_ENOSYS on an older kernel.

```
int kapi_image_preload (const char *path);
int kapi_image_unload (const char *path);
int kapi_image_list (const char *path, struct kapi_image_info *out, unsigned cap);
```

(v79) What the running kernel is: "key value" lines (name, abi, built, rev, machine, model, ram; keys may be added) -> the text's length; -KAPI_ENOSYS (and "") on an older kernel. /bin/uname.

```
int kapi_kernel_info (char *buf, unsigned cap);
```

(v80) The cores: each one's role (KAPI_CORE_*), the microseconds it was busy, an app core's owner -> 0; -KAPI_ENOSYS (out zeroed) on an older kernel. Two reads make a load.

```
int kapi_cpu_stats (struct kapi_cpu_stats *out);
```

(v80) The bytes pid's sockets received and sent, its open sockets (pid 0: every process's) -> 0; -KAPI_ENOSYS (out zeroed) on an older kernel.

```
int kapi_net_stats (int pid, struct kapi_net_stats *out);
```

(v82) A window resized by its frame (uk_win_resizable): at the release of a drag the pointer handler gets GUI_EVENT_WINRESIZE: GUI_WINRESIZE_X / _Y (the frame's new top left), _W / _H (the client area's new size) of its value; the app applies them. uikit: Root::setResizable.

```
#define GUI_EVENT_WINRESIZE 20
#define GUI_EVENT_TRAY      21  // (v95) the program's icon of the status area: value
KAPI_TRAY_OPEN (a double click) / _MENU
#define GUI_WINRESIZE_X(v)  ((int) (short) ((unsigned long long) (v) >> 48))
#define GUI_WINRESIZE_Y(v)  ((int) (short) ((unsigned long long) (v) >> 32))
#define GUI_WINRESIZE_W(v)  ((int) (((unsigned long long) (v) >> 16) & 0xFFFF))
#define GUI_WINRESIZE_H(v)  ((int) ((unsigned long long) (v) & 0xFFFF))
```

(v83) A shared library (docs/SHARED-LIBS-PLAN.md): "uikit" is SD:/lib/uikit.so, anything with a '/' or a ':' a path -> its export table (unsigned version, size; int (*init) (const TLibImports *); then its entries), mapped in this process until it ends; 0 with *err* = *-KAPI_E* (-KAPI_ENOTSUP: the library is older than min_version; -KAPI_ENOSYS on an older kernel). Apps do not call this: the library's bind object does, before main (user/Runtime/lib.h).

```
const void * kapi_lib_open (const char *name, unsigned min_version, int *err);
```

(v97) The graphics server's own library under another name (docs/POCKETUI-TECH-STUDY.md section 3): `path` opened and mapped as kapi_lib_open does (its version checked against min_version), and from now on every kapi_lib_open of `alias` -- any process's, a program's bind of "uikit" -- gets it, before a library whose real path is alias (PocketUI: kapi_lib_open_as ("SD:/lib/pocket/uikit.so", "SD:/lib/uikit.so", n, &err)). Only the process holding the graphics server's role (call it after KAPI_WS_REGISTER, before KAPI_WS_DISPLAY: every program of the session then finds the alias); both names under SD:/lib/, never

SD:/lib/appkit.so. -> its export table; 0 with *err = -KAPI_EPERM (not the server, or a name refused), -KAPI_EBUSY (alias another live process's, or the caller's on another library), -KAPI_EINVAL (path = alias), kapi_lib_open's errors, -KAPI_ENOSYS on a kernel before v97. Asked again: the same table. The alias lives while the server or a program holds the library: the server ended, it is kept for the same server started again; another server started, it is dropped.

```
const void * kapi_lib_open_as (const char *path, const char *alias, unsigned min_version, int
*err);
```

(v84) The sound's output: KAPI_SND_OUT_AUTO / _JACK / _USB / _HDMI (-1: only ask) -> what plays now, what is asked and which outputs are there (KAPI_SND_OUT_NOW / _ASKED / _HAS of the result); -KAPI_ENOSYS on an older kernel (the jack only). The Sound applet; kept in SD:/etc/sound.ini.

```
int kapi_sound_output (int out);
```

(v99) The sound's input, a microphone (KAPI_SND_IN_*): op KAPI_SND_IN_SELECT (input or -1, gain dB or -1) -> what records, what is asked, which inputs are there, the gain (KAPI_SND_IN_NOW / _ASKED / _HAS / _GAIN); _OPEN (rate) -> the rate it records at (-KAPI_EBUSY: another program records, -KAPI_ENODEV: no microphone); _READ (short *out, max) -> 16-bit mono samples taken, never waiting (-KAPI_EIO: the device is gone); CLOSE. -KAPI_ENOSYS on a kernel before v99. A program uses AudioKit's ak_in* (the rate it asks for, converted).

```
long kapi_sound_input (int op, long a0, long a1);
```

(v85) the sound's mixer: the programs that play (each has a channel), a channel's volume and mute

```
int kapi_sound_clients (struct kapi_sound_client *out, int max);
int kapi_sound_client_volume (unsigned pid, int volume, int mute);
```

(v73) Is this process protected (EL0, kern/el0.h)? Its table's memcpy is then user code, next to the table, instead of the kernel's.

```
int kapi_is_protected (void);
```

the master volume 0..10 and mute (-1: keep) -> volume | 0x100 if muted (older kernel: 10, not muted)

```
int kapi_sound_volume (int volume, int mute);
```

wpa_supplicant.conf read again + DHCP again, no reboot (-1: none / older kernel)

```
int kapi_wlan_reconnect (void);
```

v61: draws with the app's own QPU shaders (user/Libs/v3d/qpu.h builds them): gpu_program makes / replaces / frees (p = 0) a program, gpu_render2 draws batches of them (kapi_abi.h).

```
int kapi_gpu_program (int handle, const struct kapi_gpu_program *p);
int kapi_gpu_render2 (const struct kapi_gpu_frame *f, const float *v, unsigned nv, unsigned
stride,
                      const struct kapi_gpu_batch2 *b, unsigned nb, const unsigned *uni,
unsigned nuni);
```

(v62) gpu_render3: the batches' vertices where they are (kapi_gpu_batch3: off, stride), x / y framed by the kernel on the way (view: x' = v0 x + v1 w, y' = v2 y + v3 w; 0: as they are)

```
int kapi_gpu_render3 (const struct kapi_gpu_frame *f, const float *v, unsigned nfloats,
                      const struct kapi_gpu_batch3 *b, unsigned nb, const unsigned *uni,
unsigned nuni, const float *view);
```

(v63) gpu_vbuf: memory the GPU reads too -- gpu_render3 draws vertices there in place (framed in place: view 0 to draw them again; the clipped triangles into the room past nfloats) -> 0 none

```
void * kapi_gpu_vbuf (unsigned bytes);
```

(v70) gpu_texture_rect: a rectangle x, y, w x h of a texture's pixels (0xAARRGGBB) replaced, the rest kept -> 0, -1 no GPU / older kernel, -2 bad arguments. (Also v70: KAPI_GPU_F_ALPHA, a frame's target keeping its alpha; the GPU compositing service over these: user/Libs/gpucomp/gpucomp.h.)

```
int kapi_gpu_texture_rect (int handle, int x, int y, int w, int h, const unsigned *pixels,
int stride);
```

(v65) the workspaces (virtual desktops): UIKit's uk_win_desk (set, count) -> the current desk | the count << 8 | a change counter << 16, read with these:

```
#define KAPI_DESK_CUR(i)    ((i) & 0xFF)
#define KAPI_DESK_COUNT(i)  (((i) >> 8) & 0xFF)
#define KAPI_DESK_GEN(i)    (((unsigned) (i) >> 16) & 0x7FFF)
```

(v66) screen_set: the screen's resolution now (640 x 480 .. 2560 x 1600, w even); every window kept on the screen and sent GUI_EVENT_DISPLAY_RESIZE -> 0; -1 out of bounds; -2 not now (a full-screen app...); -3 the firmware refused it (old size kept); -4 an older kernel. Not kept across a reboot (SD:/cmdline.txt width= / height= are).

```
int kapi_screen_set (int w, int h);
```

(v67) Threads: more tasks in this process (its memory, window, files, sockets), preempted like it; the process ends with its main thread (tid 1). Timeouts in ms: 0 = only try, KAPI_WAIT_FOREVER = none. An older kernel: -3 / nothing (a single thread). kapi_thread_create (fn, arg, stack_size (0: 256 KB), name (0: its tid)) -> tid >= 2, -1 no memory, -2 too many (32); fn's return value is its exit code. kapi_thread_join -> 0 (*code), -1 timeout, -2 no such thread, -3 itself. The GUI belongs to the thread that pumps (the main one): a worker hands its result over with kapi_post (fn, ctx, value), which the pump (kapi_pump_events / kapi_pump_wait / kapi_wait_for_exit) runs on its thread. kapi_pump_wait (ms) sleeps until an event / a post / the close box, then pumps.

```
int kapi_thread_create (int (*fn) (void *), void *arg, unsigned stack_size, const char
*name);
void kapi_thread_exit (int code);
int kapi_thread_join (int tid, unsigned timeout_ms, int *code);
int kapi_thread_self (void);
```

Synchronisation objects (handles > 0; 256 per process). mutex: recursive, released if its owner thread ends; lock -> 0, -1 timeout, -2 bad handle. event: manual reset (stays set until reset) or auto (a wait takes it); wait -> 0, -1, -2. barrier: count threads meet; wait -> 1 for the last one in, 0 the others. kapi_sync_close frees any of them.

```
int kapi_mutex_create (void);
int kapi_mutex_lock (int h, unsigned timeout_ms);
int kapi_mutex_unlock (int h);
int kapi_event_create (int manual_reset, int initial);
int kapi_event_set (int h);
int kapi_event_reset (int h);
int kapi_event_wait (int h, unsigned timeout_ms);
int kapi_barrier_create (unsigned count);
int kapi_barrier_wait (int h);
int kapi_sync_close (int h);
```

```
int kapi_post (void (*fn) (void *ctx, long value), void *ctx, long value);
int kapi_pump_wait (unsigned timeout_ms);
```

A user-space lock between this process's threads (the allocators, newlib): an atomic swap, and a yield while another thread holds it -- nothing to create, a zeroed int is free. Not recursive. (Threads all run on core 0; the swap is still an exclusive load / store pair, since the timer may preempt a thread anywhere in its own code.) (The PC builds of the apps -- the desktop simulator, the NetSurf bench -- get the compiler's atomics and no cores.)

The core this code runs on (0: the main one; 2, 3: an app core). The kernel publishes it in TPIDRRO_EL0 (v73, kern/el0.h): every app runs at EL0, where MPIDR_EL1 may not be read (the app would be killed) -- never read MPIDR_EL1 in an app.

(on an app core -- kapi_core_run's code, which makes no kapi call -- it spins instead)

```
void kapi_lock (volatile int *l);
void kapi_unlock (volatile int *l);
```

(v68) A futex. kapi_wait_word (addr, expected, timeout_ms) sleeps while *addr == expected -> 0 woken / the value differs, 1 timeout (0 ms: only check), -1 a bad address (not 4-byte aligned, unmapped); may wake spuriously: loop on the condition. kapi_wake_word (addr) wakes its sleepers -> how many. Works across processes on a shared surface (keyed by the physical address). A word changed by an app core (no kapi there) is noticed at the next 10 ms tick. kapi_thread_priority (tid 0 self / 1 main / >= 2, prio 0 normal / 1 real time / -1 ask) -> the previous one: a real-time thread (an audio pump) runs first whenever it is ready, as long as it sleeps before its time slice ends. An older kernel: -1 / -2.

```
int kapi_wait_word (volatile unsigned *addr, unsigned expected, unsigned timeout_ms);
int kapi_wake_word (volatile unsigned *addr);
int kapi_thread_priority (int tid, int prio);
```

(v68) USB MIDI input: class-compliant devices (keyboards, interfaces), found when plugged in, even later. kapi_midi_read (ev, max) takes up to max queued events (struct kapi_midi_event: time_us, cable, status, data1, data2, device, length), oldest first, never waits -> how many; one queue for the system (256 events). kapi_midi_devices () -> attached.

```
int kapi_midi_read (struct kapi_midi_event *ev, int max);
int kapi_midi_devices (void);
```

(v69) kapi_screen_native (&w, &h): the monitor's own resolution, from its EDID -> 1, 0 unknown. kapi_set_timezone (minutes): the local time's offset from UTC at once (-720 .. 840) -> 1 ok. WIN_FLAG_FIXED (a window's flags): not movable, no minimise / maximise / close, kept centred.

```
int kapi_screen_native (int *w, int *h);
int kapi_set_timezone (int minutes);
```

The kernel's microsecond clock (CTimer::GetClockTicks: the ARM counter, same formula), the time base of kapi_midi_event.time_us. No kapi call: an app core may read it too.

```
unsigned kapi_clock_us (void);
```

(v68) Low-latency sound, for the sound owner (kapi_sound_acquire). kapi_sound_config (chunk frames 64..1024 (0: 1024), chunks ahead 1..4 (0: 4)) -> the latency now in frames ((ahead + 1) x chunk: 1024 x 4 ~116 ms by default, 256 x 2 ~17 ms), -1 not the owner / older kernel. Keep the sound_write stream shallow too (it holds up to 0.5 s): write when kapi_sound_status's free frames show little is queued. The defaults come back when the owner releases the output. kapi_sound_map () -> the mapped PCM ring

(struct kapi_sound_ring, kern/kapi_abi.h), mixed until the owner releases the output, or 0. It is plain memory: an app core (kapi_core_run) fills it with kapi_sound_ring_write, which makes no kapi call.

```
int kapi_sound_config (int chunk_frames, int ahead);
struct kapi_sound_ring * kapi_sound_map (void);
```

Frames the ring can take now.

```
unsigned kapi_sound_ring_free (const struct kapi_sound_ring *r);
```

Copy up to n s16 stereo frames into the ring -> the frames taken (no kapi call: app cores too).

```
unsigned kapi_sound_ring_write (struct kapi_sound_ring *r, const short *frames, unsigned n);
```

Reboot the machine (ABI v25). Does not return. Use to apply settings the kernel only reads at boot -- e.g. after wpaconf rewrites SD:/etc/wpa_supplicant.conf.

```
void kapi_reboot (void);
```

1 if a USB keyboard is attached & ready, else 0 (ABI v26). The `keyb` tool polls this at boot before applying a layout (it can run before USB enumeration finishes).

```
int kapi_kbd_ready (void);
```

Load a keyboard layout from a .kmap blob (ABI v27): "OKM1" + u16 rows + u16 cols + table. The kernel copies it; the caller frees `data`. 1 on success. See /etc/keymaps.

```
int kapi_set_keymap_data (const char *name, const void *data, unsigned len);
```

Hardware RNG (ABI v30): fill buf[len] with random bytes from the Pi's HW RNG. For cryptographic seeding -- the TLS entropy source uses it. Returns bytes written.

```
int kapi_random (void *buf, unsigned len);
```

Shell surfaces (ABI v35): shared 0x00RRGGBB pixel buffers for the activity-shell compositor. The shell creates one sized to a viewport, passes its id to an app; both map it (same physical frames, own VA), the app draws + presents, the shell composites.

```
int kapi_surface_create (int w, int h);
unsigned * kapi_surface_map (int id);
int kapi_surface_size (int id, int *w, int *h);
void kapi_surface_present (int id);
int kapi_surface_destroy (int id);
```

Activity-shell IPC (ABI v35): the kernel routes opaque {from_pid,type,bytes} messages between per-process mailboxes. A user compositor calls kapi_register_shell to become THE shell; apps post to it with kapi_shell_request; the shell replies / pushes async events with kapi_mailbox_send; both drain with kapi_mailbox_recv (fills *from_pid / *type, returns the payload length, or -1 if empty; blocking != 0 waits).

```
int kapi_register_shell (void);
int kapi_shell_request (int type, const void *in, unsigned len);
int kapi_mailbox_send (int target_pid, int type, const void *in, unsigned len);
int kapi_mailbox_recv (int *from_pid, int *type, void *buf, unsigned cap, int blocking);
```

(v89) The graphics server's own door to the kernel (Elegant, SD:/bin/elegant): the display, the raw input, its wait -- KAPI_WS_* (kern/kapi_abi.h) -- and the transport of UIKit's window calls (its port: KAPI_WS_CALL a request, KAPI_WS_KICK "my pixels changed", KAPI_WS_ACTIVE "is there a server"). Not

for programs: a program's windows are UIKit's uk_win_* (uikit/win.h), whoever serves them. -> >= 0, or -KAPI_Exxx (-KAPI_ENOSYS: a kernel before v89).

```
long kapi_ws_ctl (int op, long a0, long a1, long a2);
```

Memory primitives (ABI v36): the kernel's (Circle's) memset/memcpy/memmove. Real symbols (AppKit's; weak ones in a program that has the bodies inline) so the linker can see them: the freestanding app Makefiles alias the C names onto them (-Wl,--defsym,memset=kapi_memset ...), which resolves the calls GCC emits on its own (array/struct init and copies).

```
void *kapi_memset (void *dst, int c, unsigned long n);
void *kapi_memcpy (void *dst, const void *src, unsigned long n);
void *kapi_memmove (void *dst, const void *src, unsigned long n);
}
#endif
#endif
```

Friendly aliases used by the demos (create_window, present: UIKit's, uikit/win.h).

```
unsigned  get_ticks (void);
void      msleep (unsigned ms);
void      pump_events (void);
int       should_exit (void);
```

AppKit's small services (appkit_lib.inc; they were user/applib.h until 2026-10-05)

For every program, with or without a C library.

Strings. ax_strcat: src appended to dst at *pos (advanced), never past cap, always NUL-terminated. ax_app_path: "SD:apps/" (a suffix: ".app/icon.bmp"). ax_streq: 1 = the same. ax_itoa: a signed int in decimal -> its length. ax_fmt2: two digits, zero-padded, into d[0], d[1].

```
void ax_strcat (char *dst, int cap, int *pos, const char *src);
void ax_app_path (char *dst, int cap, const char *name, const char *suffix);
int  ax_streq (const char *a, const char *b);
int  ax_strlen (const char *s);
int  ax_itoa (int v, char *b);
void ax_fmt2 (char *d, int v);
```

The console: a string / a line to the standard output.

```
void ax_puts (const char *s);
void ax_putln (const char *s);
```

A minimal .ini reader: [section] headers and key=value lines, ';' or '#' comments, the spaces around trimmed. One file loaded at a time, a program (the store is the program's own); a load replaces the one before. Limits: INI_MAX entries, INI_STRLEN - 1 characters a name or a value, INI_BUFSZ - 1 bytes of file.

```
  app_ini_load (filename)      <the program's own folder>/filename -> the number of entries,
-1 absent
  app_ini_load_path (path)     any file
  app_ini_get (section, key, def), app_ini_get_int (...)   section 0 or "": the keys before
any [section]
  app_ini_count (), app_ini_section / app_ini_key / app_ini_value (i)   the entries, in the
file's order

#define INI_MAX     64
#define INI_STRLEN  64
```

```
#define INI_BUFSZ   2048
int app_ini_load_path (const char *path);   // load the .ini file at path (replacing the one loaded) -> its number of entries, -1 not opened
int app_ini_load (const char *filename);    // the same for <the program's own folder>/filename (kapi_app_dir)
```

The value of key in section (0 or "" = the keys before any [section]) of the loaded file, else def.

```
const char *app_ini_get (const char *section, const char *key, const char *def);
```

The value of key in section as a decimal integer (a sign allowed), def when the key is absent or has no digit.

```
int app_ini_get_int (const char *section, const char *key, int def);
int app_ini_count (void);   // the number of entries (key=value lines) of the loaded file
const char *app_ini_section (int i);    // the section of the entry i (0-based, in the file's order); "" out of range
const char *app_ini_key (int i);    // the key of the entry i; "" out of range
const char *app_ini_value (int i);  // the value of the entry i; "" out of range
```

The keyboard layout ("FR", "BE"...): SD:/etc/keymaps/.kmap (kapi_set_keymap_data), else a map the kernel has (kapi_set_keymap). Non-zero = done. The keyboard must be up (kapi_kbd_ready).

```
int ax_load_keymap (const char *name);
```

Starting programs (it was user/launch.h until 2026-10-05)

launch.h -- starting programs from user space. The kernel only loads ELF programs; the other formats have a RUNNER (SD:/etc/runners.ini, "extension = program"): a BASIC program (.bas, .bax) is run by SD:/bin/basic -- or an app that says it opens them: an emulator's app.txt names the games it plays, "games = Game Boy Color: gbc; Game Boy: gb" (and "opens = dol" for files it opens that are not games): installing an emulator's package is enough.

```
  lx_launch (name, args)   an app: SD:apps/<name>.app/main (an ELF), else its main.<ext>
                           run by that extension's runner (named after the app)
  lx_open (path, args)     a program file: an ELF as it is, else by its runner
  lx_runner (path, out)    the runner of a file (by its extension), 0 if none
```

C and C++ (run, init, the apps).

```
#define LX_INI  "SD:/etc/runners.ini"
char lx_low (char c);   // c in lower case ('A'..'Z' only)
int lx_len (const char *s); // the length of s (0 for a null pointer)
void lx_cat (char *d, int cap, int *n, const char *s);  // s appended to d at *n (advanced), never past cap, NUL-terminated
int lx_exists (const char *path);   // 1 if the file at path can be opened (kapi_open), else 0
```

The i-th "ext = program" line of runners.ini (0-based): 1 found, 0 past the end.

```
int lx_entry (int i, char *ext, int ecap, char *prog, int pcap);
```

Does the value of an app.txt's "games" / "opens" (the extensions after each "System:", or all the words of "opens") hold ext?

```
int lx_lists_ext (const char *v, int len, const char *ext, int games);
```

The app (SD:/apps/.app/main) whose app.txt opens files of this extension: 1 + out, 0 if none.

```
int lx_app_for (const char *ext, char *out, int cap);
```

The runner of a file, by its extension (runners.ini, else the built-in list): 1 + out, 0 if none.

```
int lx_runner (const char *path, char *out, int cap);
```

The runner's command line: "path" args (the path quoted when it has spaces).

```
void lx_cmdline (char *out, int cap, const char *path, const char *args);
```

A program file: by its runner if its extension has one, else as an ELF. name: the process name (0: the kernel's, from the path). 1 = started.

```
int lx_open_as (const char *path, const char *args, const char *name);
int lx_open (const char *path, const char *args);
```

An app bundle by folder path ("…/x.app"): its main (ELF), else main. for a runner.

```
int lx_launch_dir (const char *dir, const char *name, const char *args);
```

An app by name (SD:apps/.app), with arguments (0 / "": none). 1 = started.

```
int lx_launch (const char *name, const char *args);
```

(the tests of the kernel's table, and the PC builds: the bodies inline -- see this header's top)